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Title:

Extending nebular photometric and spectroscopic observations of Core Collapse Supernovae to the 22 magnitude range.

Abstract:

An important unsolved problem in astrophysics is how massive stars explode at the end of their lives as supernovae (SNe). The neutrino heating mechanism is still the favorite, but there are growing theoretical and observational evidences that the energy input generated by a strong magnetic field of rapidly differentially rotating protoneutron star core (magnetar) can contribute significantly. In order to investigate the possible energy input provided by the magnetar in stripped envelope core collapse SNe (SE-SNe), we propose to extend the nebular photometric observations request in the 2024B normal call to the 22 magnitude range. The latter will allow us to extend the radiative tail observation by 3 months and also complete the light curve of the SNe that will appear behind the sun at a magnitude fainter than 21 which was the magnitude limit set on the 2024B normal application.

Instruments/ telescope: VST, KASI, Speculoos

Requested observing time: VST (12 nights, minimum= 4 nights), KASI (6 nights, minimum= 2 nights), Speculoos (24 nights, 8 night).

Preferred dates: VST (1 night every month, minimum = 1 night every 3 months), KASI (Oct 23, 27, 31, Nov 28, Dec 02, and Dec 07, minimum = one night per block), Speculoos (1 nights every 10 days, minimum = 1 night every month)

Moon requirements: Grey (maximum 7 days from new moon)

SCIENTIFIC AIM AND RATIONALE

An important unsolved problem in astrophysics is how massive stars explode at the end of their lives as supernovae (SNe) (e.g. Janka 2012). The neutrino heating mechanism is still the favorite, but the energy input generated by a strong magnetic field of rapidly deferentially rotating protoneutron star core (magnetar) can contribute significantly. Such mechanism has been claimed as the main source of luminosity for superluminous SNe (SLSNe) that are explosion which are 10-100 times brighter than core collapse SNe (CCSNe). Nevertheless, recent theoretical studies have started exploring the role of magnetar energy input also in "normal" stripped-envelope CCSNe (SE-SNe) (Ertl et al., 2020), where the progenitor star has either completely (SNe Ic and SN Ib) or mostly (SN IIb) lost its hydrogen envelope prior to the explosion. For SE-SNe, in a pure neutrino driven explosion, the radioactive decay of ^{56}Ni into ^{56}Co, which has a half time of 6.1 days is the main source of the SN luminosity around maximum light. On the other hand, the radioactive decay of ^{56}Co into ^{56}Fe, which has a half time of 77.1 days, is the main source of the SN luminosity at late phases ($>$ 60 days post explosion) in the so call nebular phase. Currently the most reliable method to estimate ^{56}Ni masses uses the bolometric luminosity at nebular phases, which can be modeled as:

$$L(t) = M_{Ni}((\epsilon_{Ni} - \epsilon_{Co})e^{-t/t_{Ni}} + \epsilon_{Co}e^{-t/t_{Co}})(1 - e^{-(T_o/t)^2}) \tag{1}$$

Where t is the time since the explosion, ϵ_{Ni} and ϵ_{Co} are the specific heating rates of ^{56}Ni and ^{56}Co decay, and t_{Ni} and t_{Co} are their corresponding decay timescales. Finally, T_0 is a time scale that accounts for the fraction of energy of γ-rays and e^+ that is thermalized in the ejecta. The bolometric luminosity $L(t)$ can be estimated from multi-band photometry, therefore the only unknowns in equation (1) are M_{Ni} and T_0 that can be determined by fitting the slope and overall normalization of the radioactive tail of the light curve (see Fig. 1).

Once obtained the value of M_{Ni} from Equation (1), it is possible to estimate the bolometric SN luminosity at peak brightness through the following equation:

$$L_{peak} = \frac{2\epsilon_{Ni}M_{Ni}t_{Ni}^2}{\beta^2 t_p^2}[(1 - \frac{\epsilon_{Co}}{\epsilon_{Ni}})(1 - (1 + \beta t_p/t_{Ni})e^{-\beta t_p//t_{Ni}}) + \frac{\epsilon_{Co}t_{Co}^2}{\epsilon_{Ni}t_{Ni}^2}(1 - (1 + \beta t_p/t_{Co}))e^{-\beta t_p//t_{co}})] \tag{2}$$

Where t_p represents the time elapsed from the moment of the explosion to when the supernova reaches its peak bolometric luminosity, while β is a dimensionless parameter sensitive to several physical effects related to the explosion mechanism. Afsariardchi et al. (2022) calibrated the value of β using Dessart et al.'s (2016) models, which are based on a pure neutrino-driven explosion, obtaining $\beta = 1.125$. By entering the values of β and t_p (which is straightforward to measure from the bolometric light curve) into equation (2), one can obtain the peak bolometric luminosity of the SN in the pure neutrino driven explosions. Therefore, the eventual extra energy input required to reach the observed peak luminosity of the SN should be either due to the energy released by a central engine and/or by the possible interaction of the SN ejecta with the CSM produced by the progenitor system. However, caution should be exercised when interpreting these "excess luminosity" values as they do not account for variations from $\beta = 1.125$ that can be caused by effects such as enhanced nickel mixing and asymmetries in the explosion. Additionally, Equation (2) assumes that the radioactive component peaks at the same time as the observed light curve, which may not always be the case depending on the nature of the excess luminosity source. These shortcomings can be overcame if these excess luminosity values can be computed for a statistically significant sample and correlated with a set of SNe observables such as wideness, rise and decline time of the light curve, and the ejecta expansion velocities as measured from the minimum of the P-Cygni profiles. The latter will also help on disentangle if is the magnetar input or the CSM interaction the main source of the excess luminosity.

Summarizing, to carry out the proposed analysis we need:
(1) A well constrained estimation of the explosion time, which we can obtain from the combination of the high cadence ~ 2 days wide field surveys such the Zwicky Transient Facility (ZTF) and the Asteroid Terrestrial-impact Last Alert System (ATLAS).
(2) A well sampled early light curves that we are obtaining through our own and collaborators monitoring programs.
(3) Multiple (at least four) photometric measurements covering a wavelength range as broad as possible during the radiative tail of the light curve, (i.e., epochs >60 days). This will enable us to calculate the bolometric luminosity $L(t)$ in the nebular phase and therefore estimate M_{Ni} and T_0. We aim to obtain these data through the proposed observations. The NIR contribution to the bolometric luminosity increase its importance in the radiative tails of the SNe. NIR observations are very time consuming for objects in the magnitude range that we plan to probe, therefore, we plan to use the bolometric corrections (BC) reported in Rodríguez et al. 2023. It is worth mentioning that the uncertainties in the estimation of the SN distance will not affect our calculation of the luminosity excess, because it will impact in the same way the bolometric luminosity of the SN in both the photospheric and nebular phases.

Through this application we expand the scientific return of the observation proposed in the 2024B normal call extending for each SN the monitoring of the radiative tail for three additional months. The latter is possible when observations reach a limiting magnitude of ∼22. During 2024A 12 SE-SN suitable for the proposed observations were active in the sky reachable by the facilities we are requesting with this application. Therefore, conservatively, we propose to follow-up 10 SE-SN.

References:
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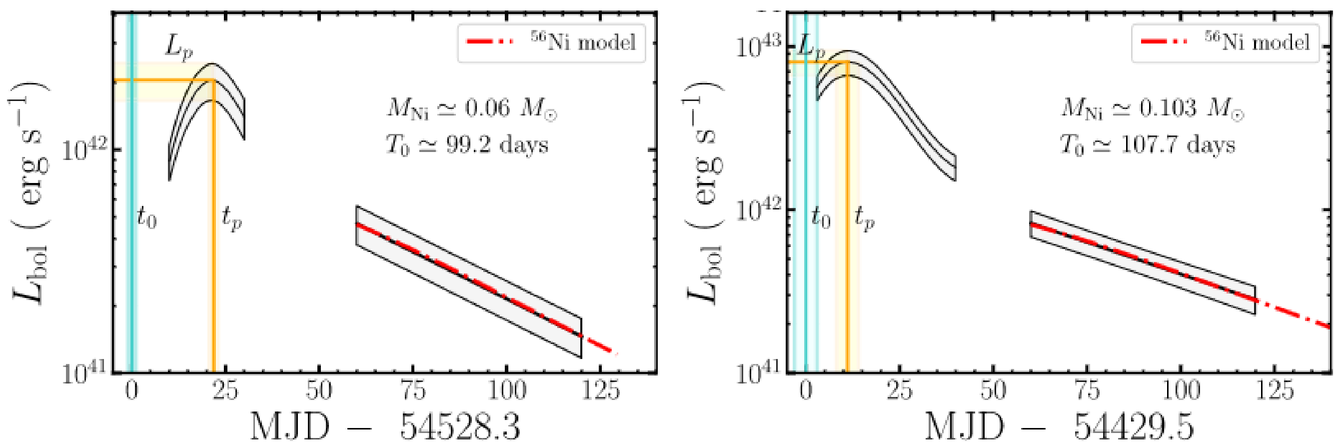

Figure 1: Bolometric light curves for SN 2008ax (left panel) and SN 2007ru (right panel). The cyan vertical line indicates the epoch of explosion, while the vertical and horizontal orange lines denote the peak time t_p and peak luminosity L_p of the bolometric light curves, respectively. The dotted-dashed red curve in the lower panels represents the best-fit ^{56}Ni model of Equation (1) to the bolometric radioactive tails. Figure adapted from Afsariardchi et al. 2022.

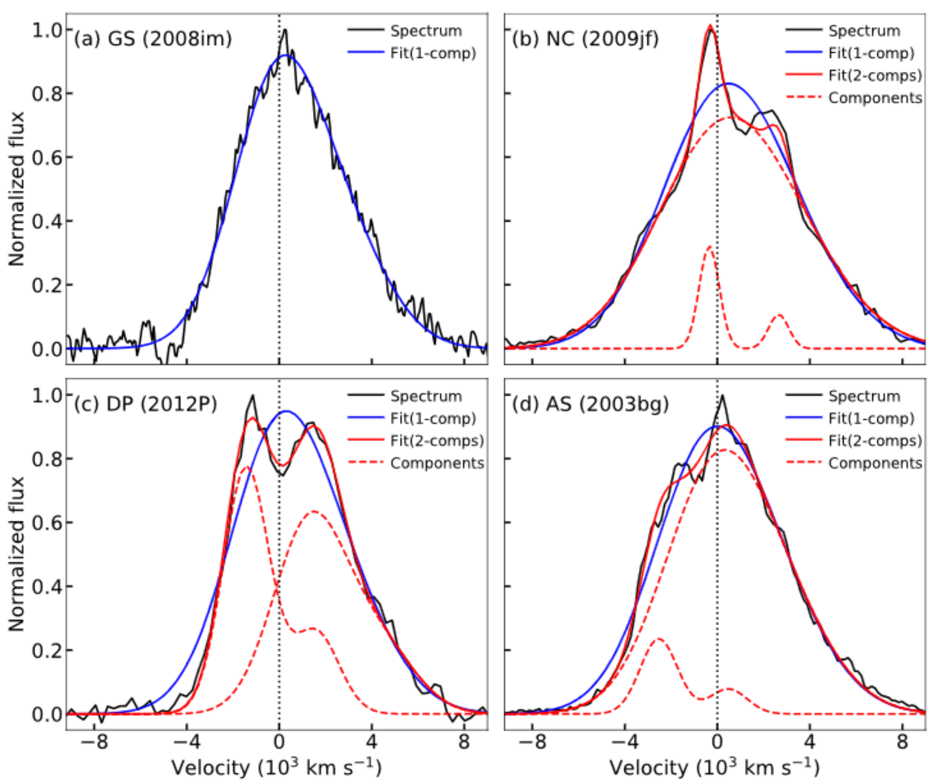

Figure 2: Examples of [OI] $\lambda\lambda$6300,6364 line profiles observable in the nebular spectra of SE-SNe. Figure adapted from Fan et al. 2022.

TECHNICAL DESCRIPTION

To extend the monitoring of the radiative tail to the 22 magnitude regime in the ugrizBV filters, we plan to use the KASI(BV), VST(uz) and Speculoos(gri) telescopes. The large filed of view of the first two telescopes is clearly an over-killing to follow-up a single object, but the diameter meet nicely the magnitude range we want to probe. For obtaining a reliable estimation of the flux of the SN we need a S/N$\sim$7.0. It is worth mentioning that usually the light of the SN is contaminated by the light of the host galaxy, therefore the background noise is usually higher than the sky noise. Using the exposure time calculator of Omegacam and assuming a moon illumination of 50% (grey time), a seeing of $1.0''$ and an airmass of 1.5, we obtain that for the u and z bands, we need 25 and 20 minutes, respectively. Therefore, to observe 10 SNe, we need (25+20) min x 10 = 450 min = 7.5 hours (one night). For KASI an exposure time calculator is not available, but base on our experience, we estimate that to reach magnitude $\sim$22th magnitude with a S/N$\sim$7.0 under the same observational conditions mentioned above, we need 30 min. for the B band and 20 min. for the V band. This means a total of $\sim$ 50 min. per SN. For Speculoos under the same sky conditions we estimate 60 min. for g,r,i.

In the case of VST there are three nights per month available to the Chilean community. We therefore request one night per month in grey or dark time. Taking into account that the present call allocate the VST time for 12 month, we request a total of 12 VST nights for the proposed observations. If it will not be possible to allocate the 12 nights, we propose to reduce the cadence to a night every three months, which results in a total request of 4 nights. During the two nights blocks when KASI is offered to the Chilean community in 2024B the average night duration is $\sim$7.5 hours during which we can observe: 7.5 x 60/50 = $\sim$9 SNe. In the case of KASI among the nights available to the Chilean community, ideally, we request the following dates: Oct 23, 27, 31, Nov 28, Dec 02, and Dec 07. In the case it will not be possible to allocate the 6 nights, we propose to remove keep only one night per block .

In the case of Speculoos the scheduling is more flexible. Considering an observations every 10 days, we request a total of 1 hour x 3 filters x 10 SNe x 18 observations = 180 hours. Considering again an average night duration of 7.5 hours, we request a total of 24 nights for the proposed observations. If it will not be possible to allocate the 24 nights, we propose to reduce the cadence to an observations every month, which results in a total request of 8 nights.

Summarizing, our request includes 12 nights of VST, 6 nights of KASI and 24 nights of Speculoos to extend the monitoring of the radiative tail to the 22 magnitude regime in the ugrizBV filters. In the case it will not possible to fully allocate the request time, we ask for a minimum of 4 nights of VST, 2 nights of KASI and 8 nights of Speculoos. Finally, in order to give maximum flexibility to our request, we note that on Omegacam are mounted also the B and V filters, thus the observations with KASI can be performed also with VST in terms of one night of KASI being equivalent of half night VST.

CURRENT STATUS OF THE PROJECT

The data obtained so far with KASI and Speculoos have been fully reduced thanks to the pipeline we developed which preforms an automatic reduction of the images collected by both telescopes including also PSF photometry.

STUDENT THESIS

These data represent a relevant contribution to the PhD thesis of Bastian Ayala. The aim of his thesis in fact put constraints of the explosion mechanism and progenitor of CCSNe studying the luminosity excess for diffract types of SE-SNe and explosion possible correlation of such excess with the degree of asymmetry in the SN ejecta.